

Somil Srivastava

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Professional Summary

Computer Science undergraduate specializing in Artificial Intelligence and Data Science with hands-on experience building Retrieval-Augmented Generation (RAG) systems, LLM-powered applications, and backend services. Skilled in Python, JavaScript, LangChain, vector databases, and REST APIs, with a strong focus on designing scalable AI workflows and production-ready software solutions.

Technical Skills

AI / GenAI: LangChain, Gemini APIs, RAG Pipelines, Embeddings, ChromaDB, Vector Search, Semantic Retrieval, Prompt Engineering, Structured Outputs

Languages & Backend: Python, JavaScript, C++, C, Node.js, Express.js, REST APIs

Databases & Tools: MongoDB, MySQL, ChromaDB, Git, GitHub, VS Code

Frontend: Streamlit, HTML, CSS

Concepts: DSA, OOP, DBMS, Software Engineering

Projects

SmartFlash – AI Study Assistant *(Python, LangChain, Gemini, ChromaDB, Streamlit)*

- Built an end-to-end Retrieval-Augmented Generation (RAG) system that ingests PDFs, performs recursive chunking, generates embeddings, and stores vectors in ChromaDB for semantic retrieval.
- Developed a topic-aware learning engine that retrieves relevant document context and generates AI-powered flashcards and multiple-choice questions using Gemini 2.5 Flash.
- Implemented semantic search over embedded documents, enabling users to generate study material from natural-language topics and chapter-level queries.
- Designed a modular LangChain pipeline for document loading, chunking, retrieval, prompt orchestration, and structured content generation using Pydantic schemas.

Agentic Content Optimization Workflow *(Python, LangGraph, Gemini 2.5 Flash, Pydantic)*

- Built a LangGraph StateGraph workflow for generation, evaluation, and iterative refinement of AI-generated content.
- Implemented an LLM-as-a-Judge system using Pydantic structured outputs for quality assessment.
- Designed conditional routing that accepted outputs or triggered reflection-based improvements.
- Maintained workflow context through typed state management, ensuring reliable information flow across graph nodes.
- Built a feedback-driven optimization loop where evaluator outputs were incorporated into subsequent prompts for content improvement.
- Orchestrated Gemini and Llama models for generation, evaluation, and refinement tasks within a unified workflow.

Health.IO – AI-Powered Healthcare Platform *(Node.js, Express.js, MongoDB, Groq APIs)*

- Built an AI-powered healthcare platform integrating Groq-hosted LLMs for symptom analysis and conversational assistance.
- Designed REST APIs and prompt-engineered workflows to generate contextual health insights from user-reported symptoms.

Education

Pranveer Singh Institute of Technology, Kanpur

B.Tech in Artificial Intelligence and Data Science CGPA: 7.4

Dec 2022 – Jun 2026

Kanpur, India

Durga Prasad Vidya Niketan, Kanpur

AISSCE (Class XII) Percentage: 72.6%

Mar 2021 – Apr 2022

Kanpur, India

Achievements & Certifications

- **Finalist (Top 24 Nationwide)** at PEC HACKS 2024 for developing and presenting an AI-powered healthcare platform.
- Solved **500+ DSA problems** across competitive programming platforms focused on algorithms and backend engineering.
- Achieved **5-Star HackerRank rating** and earned multiple **LeetCode badges** in Algorithms and DSA study plans.
- **Python Certification** – HackerRank (2024).